# Analytical Expressions of the Guided Mode Field Distribution: Analysis and Comparison

Ning Hua Zhu, Yi Kun Lin, Zheng De Wu, and Weigan Lin

*Abstract*—**This letter presents methods for determining the parameters of the three different approximate solutions for the fundamental mode field distribution of Ti:LiNbO$_3$ optical waveguides by using the variational technique. This analysis gives closer approximation not only to the field distribution, but also to the propagation constant. Comparison with the exact results for the exponential index distribution in depth shows that these trial functions are closer to the real solution. It is also shown that these solutions are as accurate as the effective index method for Ti:LiNbO$_3$ stripe waveguides. The analysis provides an approach for comparing and assessing these solutions.**

## I. Introduction

The analytical expression of the guided mode field components of Ti:LiNbO$_3$ optical waveguides is attractive for designing the integrated optical devices [1], [2] and achieving low fiber–waveguide–fiber insertion loss [3], [4]. However, an exact analytical solution of the mode profile has not been established until now. Although the finite-element method can give an exact numerical solution for this problem [5], it is not suited for the optimization procedures because a great deal of calculation time is required. Several approximate solutions for the predominant field distribution have been reported. McCaughan and Murphy [6] used a symmetric Gaussian and two half-Gaussians to describe the width and depth mode profile, and the mode sizes were obtained from measurement. Marcuse [2] approximated the $x$ and $y$ dependence of the transverse field component of the fundamental mode by Gaussian and Hermite–Gaussian functions. The mode dimensions and the effective index were calculated directly from controllable fabrication parameters by employing a variational analysis [7]. Strake *et al.* [5] obtained the analytical expression of the scalar wave equation based on the effective index method. The parameters of the solution were chosen by the point matching technique and the comparison with the FEM results.

In this letter, we use the variational method to choose the parameters of the trial solutions from the diffusion parameters. The computer simulations for different Ti stripe width are presented. The comparison with the effective index method shows that the analytical solutions are accurate enough for the optimization and design of the integrated optical devices.

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## II. Analytical Expressions of the Field Distributions

### A. Refractive-Index Profile

For a Ti:LiNbO$_3$ channel waveguide with the initial width of the Ti stripe $W$, the refractive index profile can be expressed as [8]

$$n^2(x, y) = \begin{cases} 1, & y < 0 \\ n_b^2 + (n_s^2 - n_b^2) f(y/D_y) g(2x/W) & y > 0 \end{cases} \tag{1a}$$

where

$$f(y/D_y) = \exp(-y^2/D_y^2) \tag{1b}$$

$$g(2x/W) = \tfrac{1}{2}\{\mathrm{erf}[W/2D_x(1 + 2x/W)] + \mathrm{erf}[W/2D_x(1 - 2x/W)]\} \tag{1c}$$

$n_b$ is the refractive index of the substrate, $n_s$ is the refractive index at $x = y = 0$ for a sufficiently wide Ti stripe and $D_x$ and $D_y$ are the diffusion width and depth, respectively.

### B. Expression of Half-Gaussians

The field distribution of the fundamental mode is approximated in [6] to a symmetric Gaussian in the width direction and two half-Gaussians in the depth direction. The mode profile is then written as

$$\phi(x, y) = \begin{cases} C \exp[-x^2/a_1^2 - (y - d)^2/a_2^2] & y \le d \\ C \exp[-x^2/a_1^2 - (y - d)^2/a_3^2] & y \ge d \end{cases} \tag{2}$$

where $d$ is the $y$ position of the field maximum and $C$ is the normalization coefficient. Here, the four parameters $a_1$, $a_2$, $a_3$, and $d$ are found by the variational method. If $\phi(x, y)$ is a good description of the guided mode, it can be used to find the propagation constant $\beta$ from equation

$$\beta^2 = \frac{\iint \phi \nabla_t^2 \phi \, dS + k^2 \iint n^2 \phi^2 \, dS}{\iint \phi^2 \, dS}. \tag{3}$$

We substitute the trial solution into the above equation and obtain

$$\beta^2 = k^2 n_b^2 + \frac{8(n_s^2 - n_b^2)k^2}{a_1(a_2 + a_3)\pi} QR + \frac{k^2(1 - n_b^2)a_2}{a_2 + a_3} \cdot \text{erfc}\,(\sqrt{2}\, d/a_2) - \frac{a_1^2 + a_2 a_3}{a_1^2 a_2 a_3} \tag{4a}$$

where

$$Q = \int_0^\infty g(2x/W) \exp(-2x^2/a_1^2)\, dx \tag{4b}$$

$$R = \frac{\sqrt{\pi}}{2\beta_2} \exp(-\alpha_2\, d/D_y^2) \{\text{erf}\,[\beta_2(d - \alpha_2)] + \text{erf}\,(\beta_2\alpha_2)\} + \frac{\sqrt{\pi}}{2\beta_3} \exp(-\alpha_3\, d/D_y^2) \cdot \text{erfc}\,[\beta_3(d - \alpha_3)] \tag{4c}$$

$$\beta_i = \sqrt{a_i^2 + 2D_y^2}/a_i D_y \tag{4d}$$

$$\alpha_i = 2\, d/a_i^2 \beta_i^2 \qquad i = 2, 3. \tag{4e}$$

The optimal values of $a_1$, $a_2$, $a_3$, and $d$ correspond to the largest value of $\beta$.

### C. Gaussian and Hermite-Gaussian Approximations

For the weakly guided optical waveguides, Marcuse expressed the optical field of the lowest order modes as [2]

$$\phi(x, y) = \frac{4y}{\sqrt{W_x W_y^3 \pi}} \exp(-x^2/W_x^2 - y^2/W_y^2). \tag{5}$$

The newly defined parameters $W_x$ and $W_y$ are $\sqrt{2}$ times less than the old ones [2], but this does not make any difference to the propagation constant and the field distributions. So we still follow the former definition. A procedure similar to the above section is applied to determine $W_x$ and $W_y$. The variational equation for the propagation constant is

$$\beta^2 = k^2 n_b^2 + \frac{4(n_s^2 - n_b^2)k^2}{\sqrt{2\pi}\, W_x} \left(1 + \frac{W_y^2}{2D_y^2}\right)^{-(3/2)} \cdot Q - \frac{1}{W_x^2} - \frac{3}{W_y^2} \tag{6a}$$

where

$$Q = \int_0^\infty g(2x/W) \exp(-2x^2/W_x^2)\, dx. \tag{6b}$$

### D. Extended Effective Index Method

In order to obtain the analytical solution of the field distribution in the $y$ direction, the following function is used instead of (1b)

$$\varphi_y(u) = 1 - Mth^2(\eta u) \tag{7}$$

where $u = y/D_y$, $M$ and $\eta$ are the parameters to be determined. The local effective index of the refraction is therefore expressed as

$$n_{\text{eff}}^2(s) = n_b^2 + (1 - M)g(s)(n_s^2 - n_b^2) + (\eta/\alpha_y k D_y)^2 (2q + 1 + \mu'(s))^2 \tag{8a}$$

where

$$\mu'(s) = \tfrac{1}{2}[1 - \sqrt{1 + 4M(n_s^2 - n_b^2)g(s)(\alpha_y k D_y/\eta)^2}] \tag{8b}$$

and $s = 2x/W$. Similarly, the analytical expression of the field distribution in the $x$ direction can be obtained by matching following two functions

$$\Psi(s) = \frac{n_{\text{eff}}^2(s) - n_b^2}{n_{\text{eff}}^2(0) - n_b^2} \tag{9a}$$

$$\varphi_x(s) = 1 - Rth^2(\kappa s). \tag{9b}$$

By properly choosing the parameters $M$, $\eta$, $R$, and $\kappa$, a good approximation can be achieved. The mode effective index is given by

$$N_{\text{eff}}^2 = n_b^2 + (1 - R)[n_{\text{eff}}^2(0) - n_b^2] + (2\kappa/\alpha_x k W)^2 (p + \mu)^2 \tag{10a}$$

where

$$\mu = \tfrac{1}{2}[1 - \sqrt{1 + 4R(\alpha_x k W/2\kappa)^2 (n_{\text{eff}}^2(0) - n_b^2)}] \tag{10b}$$

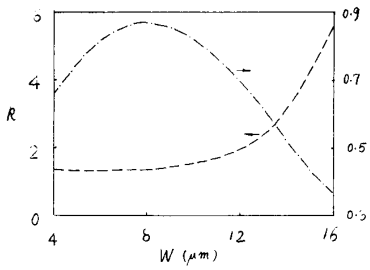
$p$ and $q$ are the mode orders in the $x$ and $y$ directions, respectively, $\alpha_x$ and $\alpha_y$ represent the crystal anisotropy. The transverse field component is then written as

$$\Phi(x, y) = N_{pq} X_p(s) Y_q(u; s) = N_{pq}\, ch^{\mu}(\kappa s)\, C_p^{\mu}[-ish(\kappa s)] \cdot ch^{\mu'}(\eta u)\, C_{2q+1}^{\mu'}[-ish(\eta u)] \tag{11}$$

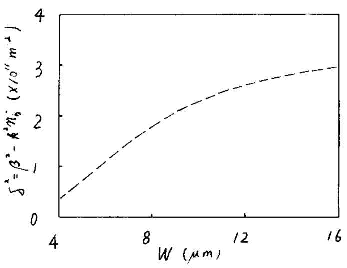
where $C_p^{\mu}$ and $C_{2q+1}^{\mu'}$ are Gegenbauer polynomials. For the well-guided modes, the accuracy of this method depends on the determination of the four parameters $M$, $\eta$, $R$, and $\kappa$. Different matching methods will bring about quite different results. Here, $M$ and $\eta$ are chosen in such a way that the following overlap integral reaches its peak value.

$$T = \frac{\left| \int_0^t \varphi_y f\, dy \right|^2}{\int_0^t \varphi_y^2\, dy \int_0^t f^2\, dy}. \tag{12}$$

In this formula $t$ satisfies $\varphi_y(t) = 0.001$. The optimal values $M = 1.038$ and $\eta = 1.061$ are found and the corresponding value of $T$ is over 0.999. The same procedure is employed for the choice of $R$ and $\kappa$. Fig. 1 shows computer simulations for different Ti stripe width. For stripe width of 4–16 $\mu$m, the overlap integral value is over 0.997 and this large value near unity implies that the approximation is appropriate.




(a)




(b)

Fig. 1. Parameters and the propagation constants obtained by matching technique through overlap integral.

In order to compare this solution with the above mentioned solutions, we have used the variational method to this expression. According to the assumption that $Y(u; s)$ is a slowly varying function of $x$ [9], $\mu'$ is treated as a parameter in the analysis. Substituting the fundamental mode solution into (3) and setting $\alpha_x = \alpha_y = 1$ (i.e., neglecting the anisotropy of the crystal), we obtain

$$\beta^2 = k^2 n_b^2 + \frac{4(n_s^2 - n_b^2)k^2}{B(1.5, -\mu' - 1)B(0.5, -\mu)} \cdot \int_0^\infty e^{-\zeta^2/\eta^2} ch^{2\mu'}(\zeta) sh^2(\zeta)\, d\zeta \cdot \int_0^\infty g(\xi/\kappa) ch^{2\mu}(\xi)\, d\xi + Z \tag{13a}$$

where

$$Z = \frac{(3\mu' + 1)\eta^2}{D_y^2} + \frac{4\mu\kappa^2}{W^2} + \frac{\mu'(\mu' - 1)\eta^2 B(2.5, -\mu' - 1)}{D_y^2 B(1.5, -\mu' - 1)} + \frac{4\mu(\mu - 1)\kappa^2 B(1.5, -\mu)}{W^2 B(0.5, -\mu)}. \tag{13b}$$

$B(p, q)$ is the complete beta function.

## III. Simulation Results

In order to investigate the accuracy of the above methods, we first consider the one-dimensional optical waveguide with an exponential index distribution [10]

$$n = \begin{cases} 1, & y < 0 \\ n_1 + \Delta n \exp(-y/d) & y > 0 \end{cases}. \tag{14}$$

According to (2), (5), and (11), the field profile of the fundamental mode can be described by following three trial functions

$$\phi_a = \begin{cases} C \exp[-(y - b)^2/a_1^2] & y \le 0 \\ C \exp[-(y - b)^2/a_2^2] & y \ge 0 \end{cases} \tag{15}$$

$$\phi_b = C ch^{\mu'}(\eta y/d) sh(\eta y/d) \tag{16}$$

$$\phi_c = C y \exp(-y^2/W^2) \tag{17}$$

where $C$ is the normalization constant, $a_1$, $a_2$, $b$, $\mu'$, $\eta$, and $W$ are parameters to be determined. Substituting the above equations into (3), we obtain the following variational equations for the propagation constant

$$\beta_a^2 = k^2 n_1^2 + \frac{a_1(1 - n_1^2)k^2}{a_1 + a_2} \operatorname{erfc}(\sqrt{2}\, b/a_1) + \frac{2\, \Delta n n_1 k^2}{a_1 + a_2} Q - \frac{1}{a_1 a_2} \tag{18a}$$

$$Q = a_1 \exp\left(\frac{a_1^2}{8\, d^2} - \frac{b}{d}\right)\left[\operatorname{erf}(\sqrt{2}\, a_1/4\, d) + \operatorname{erf}\left(\sqrt{2}\left(\frac{b}{a} - \frac{a_1}{4\, d}\right)\right)\right] + a_2 \exp\left(\frac{a_2^2}{8\, d^2} - \frac{b}{d}\right) \operatorname{erfc}(\sqrt{2}\, a_2/4\, d) \tag{18b}$$

$$\beta_b^2 = k^2 n_1^2 + \frac{(3\mu' + 1)\eta^2}{d^2} + \frac{\mu'(\mu' - 1)\eta^2 B(2.5, -\mu' - 1)}{d^2 B(1.5, -\mu' - 1)} + \frac{4\, \Delta n n_1 k^2 F}{B(1.5, -\mu' - 1)} \tag{19a}$$

$$F = \int_0^\infty \exp(-\xi/\eta) ch^{2\mu'}(\xi) sh^2(\xi)\, d\xi \tag{19b}$$

$$\beta_c^2 = k^2 n_1^2 + \frac{8\, \Delta n n_1 k^2}{\sqrt{\pi}} \cdot \int_0^\infty \exp(-\xi^2 - \xi W/\sqrt{2}\, d)\xi^2\, d\xi - \frac{3}{W^2}. \tag{20}$$

The subscripts $a$, $b$, and $c$ in (15)–(20) are related. For the example discussed in [10], where $n_1 = 2.47$, $\Delta n = 0.06$, $d = 2.5$ μm, and $\lambda = 0.6328$ μm, the calculated results are summarized in Table I, and the corresponding normalized field distributions are plotted in Fig. 2. We see that the Hermite–Gaussian is lower than the exact solution both at the surface and into the substrate. For the Gaussian dependence of the refractive index profile in

TABLE I
CALCULATED PARAMETERS AND PROPAGATION CONSTANT FOR THE WAVEGUIDE WITH THE EXPONENTIAL INDEX PROFILE IN DEPTH

| Solution | Parameters | $\beta$ ($\times 10^8$ m$^{-1}$) |
|---|---|---|
| $\phi_a$ | $a_1 = 0.290$ $\mu$m, $a_2 = 1.109$ $\mu$m, $b = 0.420$ $\mu$m | 0.24891813 |
| $\phi_b$ | $\mu' = -4.165$, $\eta = 2.059$ | 0.24893376 |
| $\phi_c$ | $W = 0.977$ $\mu$m | 0.24892758 |
| Exact Solution [10] | | 0.24899183 |

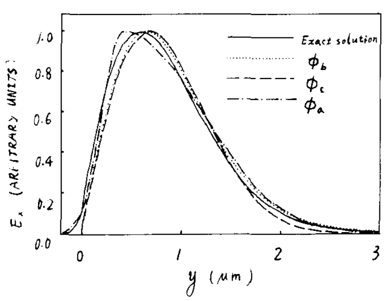



Fig. 2. Fundamental mode profiles in the 1-D optical waveguide with the exponential index distribution.

TABLE II
CALCLATED PARAMETERS ($\mu$ AND $\mu'$) AND CORRESPONDING PROPAGATION CONSTANT VERSUS $\kappa$ IN CASE $\eta = 1.06$ AND $W = 6$ $\mu$m

| $\kappa$ | $\mu$ | $\mu'$ | $\delta^2$ ($\times 10^{11}$ m$^{-2}$) |
|---|---|---|---|
| 0.4 | −6.118 | −2.597 | 0.9999129 |
| 0.5 | −4.047 | −2.597 | 1.0011063 |
| 0.6 | −2.914 | −2.597 | 1.0017446 |
| 0.7 | −2.253 | −2.597 | 1.0016496 |
| 0.8 | −1.798 | −2.597 | 1.0007878 |

depth, the same conclusion can be drawn from the experiment results [4]. $\phi_a$ gives a good approximation for the field near the dielectric-air interface. From the above analyses it follows that the trial function $\phi_b$ is closer to the exact solution, but for the practical applications these three approximate solutions are accurate enough.

Now, we consider a Z-cut $LiNbO_3$ strip waveguide having the following parameters $D_x = 3.6$ $\mu$m, $D_y = 4.0$ $\mu$m [11], $\Delta n = 0.01$, and $\lambda = 1.32$ $\mu$m. For the guided modes, $\beta^2$ is slightly larger than $k^2 n_b^2$, so we use $\delta^2 = \beta^2 - k^2 n_b^2$ instead of $\beta^2$. The results of the extended effective index method for $W = 6$ $\mu$m are listed in Tables II and III. For the given $\eta$ and $\kappa$, we can obtain almost the same maximum value of $\delta^2$ by adjusting $\mu'$ and $\mu$. This property makes it difficult to find the optimal parameters. Since the different combination of $\eta$ and $\mu'$ or $\kappa$ and $\mu$ does not make much difference to $\delta^2$, the parameters $\eta$ and $\kappa$ can be set to be constant. According to Tables II and III, $\eta$ and $\kappa$ are chosen to be 1.06 and 0.6, respectively.

By using the procedures of the above section, we find the parameters for all of the trial solutions and the corresponding propagation constants, as shown in Fig. 3. Plot-

TABLE III
CALACULATED PARAMETERS ($\mu$ AND $\mu'$) AND CORRESPONDING PROPAGATION CONSTANT VERSUS $\eta$ IN CASE $\kappa = 0.6$ AND $W = 6$ $\mu$m

| $\eta$ | $\mu$ | $\mu'$ | $\delta^2$ ($\times 10^{11}$ m$^{-2}$) |
|---|---|---|---|
| 0.94 | −2.914 | −3.021 | 1.0003861 |
| 1.00 | −2.914 | −2.791 | 1.0014852 |
| 1.06 | −2.914 | −2.597 | 1.0017446 |
| 1.12 | −2.914 | −2.436 | 1.0010749 |
| 1.18 | −2.914 | −2.297 | 0.9993434 |

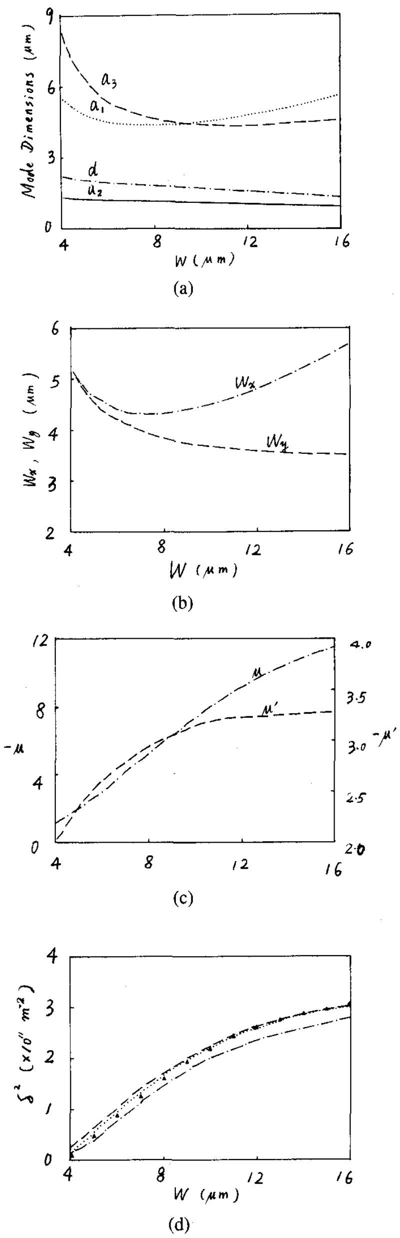



Fig. 3. (a), (b), and (c) Parameters of three different trial solutions calculated by the variational method. (d) The corresponding propagation constants, compared with those of EIM. Dash-dotted curve: expression of half-Gaussians; dotted curve: Gaussian and Hermite–Gaussian approximation; dashed curve: extended EIM; closed triangle: EIM.

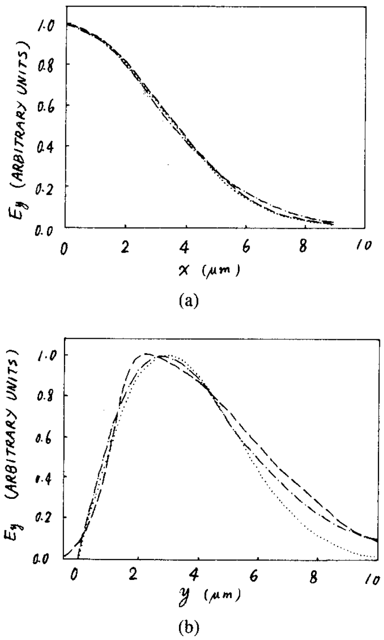



Fig. 4. Normalized field distribution in (a) the width and (b) depth directions for $W = 6$ μm. Dashed curve: expression of half-Gaussians; dash-dotted curve: extended EIM; dotted curve: Gaussian and Hermite–Gaussian approximation.

ted in Fig. 4 are the normalized field distributions. It can be seen that all three solutions give nearly the same description in the $x$ direction, but that the profiles in the $y$ direction are slightly different. Fig. 3(d) also includes the results of the effective index method [8]. The above discussions follow that the solution approximated by half-Gaussians is less accurate than the other two trial solutions, but it gives a more accurate description for the field distribution near the substrate surface. Considering that $\beta^2$ is much larger than $\delta^2$ and comparing with the results of EIM, we may conclude that these approximate expressions are accurate enough for designing and optimizing the optical waveguide devices.

## IV. Conclusion

In this letter, we have given the detailed procedures of determining the parameters of the three different analytical expressions of the field distributions in diffused channel waveguides by using the variational method. The advantage of this method is that the propagation constant can be approximately obtained simultaneously, i.e., both the amplitude and argument of the transverse field component can be found. The proposed methods were employed to analyze the waveguide with exponential index distribution in depth. We also compared the propagation constants obtained by the variational method with that of the effective index method for the $LiNbO_3$ strip waveguide. The results show that the three approximate solutions are quite accurate. Although (5) is closer to the real solution, the optimum parameters are difficult to obtain. There are only two parameters to be determined in Marcuse's trial solution. The others have four unknown parameters and are slightly more complicated in mathematical style. Another advantage of Marcuse's solution besides its simplicity is high accuracy. So, it is well suited for the design and optimization of integrated optical devices.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